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Filter and filter assembly for paint overspray capture

Abstract

The disclosed technology includes an air filter box for filtering particulates from air. The air filter box can include a filter media pack having a filter layer and a plurality of filter media supports configured to support the filter layer. The air filter box can also include a filter housing having an inlet, an outlet, and a plurality of support tabs configured to align and support the filter media pack inside of the filter housing. The filter media pack can be configured to be folded and placed inside of the housing for cooperative engagement with the support tabs. The filter box can be configured to channel a mixture of air and particulates from the inlet, through the filter media pack, and out the outlet to create filtered air.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of, and claims priority under 35 U.S.C. § 120 to, U.S. Non-Provisional patent application Ser. No. 18/223,603 filed Jul. 19, 2023, which is a continuation of U.S. Non-Provisional patent application Ser. No. 16/952,792 filed Nov. 19, 2020 which claims priority to U.S. Provisional Patent Application No. 62/938,165 filed 20 Nov. 2019 and U.S. Provisional Patent Application No. 63/032,123 filed 29 May 2020, the contents of which are incorporated herein by reference in their entirety as if set forth verbatim.

FIELD OF TECHNOLOGY

(1) The present disclosure relates generally air filters and filter assemblies for filtering particulates from air and, more particularly, for filtering paint particulates from air.

BACKGROUND

(2) For centuries paint has been used to protect and beautify materials. Traditionally, paint is applied to a surface using a paintbrush, roller, or other device which soaks up the paint and then applies it to the surface by physical contact. However, a more recent method of painting utilizes compressed gasses (e.g., compressed air, propane, butane, isobutane, methyl ethyl, nitrous oxide, or carbon dioxide, etc.) to create an aerosol mixture of paint particles and gas to apply paint to a surface. The resultant aerosol mixture is propelled from a spray device and the paint particles are applied to the surface in a constant stream allowing the user to rapidly apply paint to a surface. Although this form of painting may save time, and potentially create better finishes, there are dangers associated with using this painting method including inhaling toxic fumes and paint particulates.

(3) To reduce the danger of inhaling toxic fumes and paint particulates, various methods of filtering out the toxic particulates from the air have been developed. For example, air filtration systems help to reduce harmful particulates in the air. Furthermore, designated paint booths can be used to isolate the area where paint fumes may exist from other areas where one would expect clean air. Air filtration systems are used to provide a fresh supply of breathable air to the painters inside the paint booth and filter out particulates from air exiting the paint booth.

(4) To capture the harmful fumes and particulates exiting a paint booth, various types of filters are used. One type of filter comprises a rectangular (e.g., a square) box-shaped filter which has filter media placed inside of the filter box to capture paint particulates and allow filtered air to exit to the atmosphere or be recirculated through the air filtration system. The box filter can be disposed once it has reached the end of its usable life. Many current designs of box filters utilize a series of cardboard baffles or shredded paper for filter media. However, these designs tend to prematurely clog up the entrance of the filter rather than utilize the entire filter media available leading to an ineffective use of the filter media. Furthermore, these designs are cumbersome to assemble causing the user frustration and increasing the margin for error in the assembly of the filter. Therefore, the present invention is directed to an improved paint box filter using alternative filter media which can effectively capture particulates and reduce the amount of time required to assemble the paint filter.

SUMMARY

(5) The disclosed technology relates generally to filters and filter assemblies for capturing paint overspray.

(6) The disclosed technology can include an air filter box for filtering particulates from a mixture of air and particulates. The air filter box can include a filter media pack configured to receive a mixture of air and particulates and filter at least a portion of the particulates from the mixture of air and particulates to create filtered air. The filter media pack can include a filter layer configured to filter at least a portion of the particulates from the mixture of air and particulates and a plurality of filter media supports configured to provide support to the filter layer.

(7) The filter box can also include a filter housing configured to receive and support the filter media pack. The filter housing can include an inlet configured to receive the mixture of air and particulates, an outlet configured to receive the filtered air from the filter media pack and release the filtered air out the outlet, and a plurality of support tabs configured to align and support the filter media pack inside of the filter housing. The filter media pack can be configured to be folded and placed inside of the filter housing for cooperative engagement with the support tabs. Furthermore, the air filter box can be configured to channel the mixture of air and particulates from the inlet, through the filter media pack, and out the outlet to create the filtered air.

(8) The filter layer can be a first filter layer configured to filter at least a first portion of the particulates from the mixture of air and particulates and the air filter box can further include a second filter layer affixed to the first filter layer that can be configured to filter at least a second portion of the particulates from the mixture of air and particulates. The first filter layer can be thermally bonded to the second filter layer.

(9) The plurality of filter media supports can be located between the first filter layer and the second filter layer. The plurality of filter media supports can also include a plurality of perforations configured to channel the mixture of air and particulates through the first filter layer and the second filter layer at the plurality of perforations to create the filtered air. Alternatively, or in addition, the plurality of filter media supports can form a gap between adjoining filter media supports such that the plurality of filter media supports channel the mixture of air and particulates through the first filter layer and the second filter layer at the gap to create the filtered air.

(10) The filter media pack can be configured to form a friction fit seal at an interface between the filter media pack and the filter housing. The friction fit seal can be configured to prevent the mixture of air and particulates from being directed around an outer edge of the filter media pack.

(11) The filter housing can be configured to be folded from a relatively flat state to a deployed state. The plurality of support tabs can include inlet guide fingers configured to support and align the filter media pack inside of the filter housing. The inlet guide fingers can be configured to fold inwardly from a top and bottom of a front outer wall of the filter housing to, at least partially, form the inlet. The plurality of support tabs can include inlet support tabs configured to align and support the filter media pack inside of the filter housing. The inlet support tabs can be configured to fold inwardly from a side of a front outer wall of the filter housing to, at least partially, form the inlet. The plurality of support tabs can include outlet guide fingers configured to support and align the filter media pack inside of the filter housing. The outlet guide fingers can be configured to fold inwardly from a top and bottom of a rear outer wall of the filter housing to, at least partially, form the outlet. The plurality of support tabs can include outlet support tabs configured to support the filter media pack inside of the filter housing. The outlet support tabs can be configured to fold inwardly from a side of a rear outer wall of the filter housing to, at least partially, form the outlet. The filter layer can include lofted polyester.

(12) The disclosed technology can include an air filter assembly. The air filter assembly can include a plurality of air filter boxes. Each air filter box can include a filter media pack configured to receive a mixture of air and particulates and filter at least a portion of the particulates from the mixture of air and particulates to create filtered air. The filter media pack can include a filter layer configured to filter at least a portion of the particulates from the mixture of air and particulates. The filter media pack can include a plurality of filter media supports configured to provide support to the filter layer.

(13) Each filter box can further include a filter housing configured to receive and support the filter media pack. The filter housing can include an inlet configured to receive the mixture of air and particulates and an outlet configured to receive the filtered air from the filter media pack and release the filtered air out the outlet. The filter media pack can be configured to be folded and placed inside of the filter housing. The air filter box can be configured to channel the mixture of air and particulates from the inlet, through the filter media pack, and out the outlet to create the filtered air.

(14) The air filter assembly can include a filter assembly housing configured to receive and support the plurality of air filter boxes. The filter assembly housing can include an air filter assembly inlet configured to receive the mixture of air and particulates and channel the mixture of air and particulates toward the plurality of air filter boxes. The filter assembly housing can include an air filter assembly outlet configured to receive the filtered air from the plurality of air filter boxes and channel the filtered air out the air filter assembly outlet. The filter assembly housing can be configured to channel the mixture of air and particulates through the plurality of the air filter boxes simultaneously.

(15) The air filter assembly can include a first air filter box that can be offset a distance from a second air filter box such that a distance between the inlet of the first air filter box and a wall of the filter assembly housing is greater than a distance between the inlet of the second air filter box and the wall of the filter assembly housing. The first air filter box can be nearer the filter assembly inlet

than the second air filter box.

(16) The air filter assembly inlet can include a sloped edged configured to channel the mixture of air and particulates toward the plurality of air filter boxes. The air filter assembly can include one or more air deflectors configured to channel the mixture of air and particulates through the plurality of air filter boxes. The air filter assembly can include reinforcing tabs configured to receive a restraint device of an air filtration system. The air filter assembly inlet can be located proximate a top surface of the air filter assembly and the air filter assembly outlet can be located proximate a back surface of the air filter assembly.

(17) Additional features, functionalities, and applications of the disclosed technology are discussed herein in more detail.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate multiple examples of the presently disclosed subject matter and serve to explain the principles of the presently disclosed subject matter. The drawings are not intended to limit the scope of the presently disclosed subject matter in any manner.

(2) FIG. 1 illustrates a top perspective view of an example filter media pack, in accordance with the disclosed technology.

(3) FIG. 2A illustrates a top view of an example filter media pack, in accordance with the disclosed technology.

(4) FIG. 2B illustrates a side view of an example filter media support, in accordance with the disclosed technology.

(5) FIG. 2C illustrates a side view of an example filter media support, in accordance with the disclosed technology.

(6) FIG. 3 illustrates a top view of an example filter box, in accordance with the disclosed technology.

(7) FIG. 4A illustrates a front perspective view of an example filter box, in accordance with the disclosed technology.

(8) FIG. 4B illustrates a rear perspective view of an example filter box, in accordance with the disclosed technology.

(9) FIG. 5A illustrates a front view of an example filter box, in accordance with the disclosed technology.

(10) FIG. 5B illustrates a rear view of an example filter box, in accordance with the disclosed technology.

(11) FIG. 6 illustrates a top view of a filter housing in an unassembled state, in accordance with the disclosed technology.

(12) FIG. 7 illustrates a side view of an example air filter assembly, in accordance with the disclosed technology.

DETAILED DESCRIPTION

(13) The present disclosure relates generally to filters and filter assemblies for filtering particulates from air and, more particularly, for filtering paint particulates from air. The disclosed technology, for example, can include an air filter box having a filter media pack and a filter housing. The filter media pack can be placed inside of the filter housing to form the air filter box and used to filter particulates from a mixture of air and particulates to create filtered air. For example, the filter box can be used with an air filtration system of a paint booth to filter paint particulates that often result from paint overspray during the painting process. The mixture of air and particulates can be passed through the air filter by the air filtration system to create filtered air. In some examples, more than

one filter box can be assembled in a housing to form a filter assembly having multiple filter boxes.

(14) Although certain examples of the disclosed technology are explained in detail, it is to be understood that other examples, embodiments, and implementations of the disclosed technology are contemplated. Accordingly, it is not intended that the disclosed technology is limited in its scope to the details of construction and arrangement of components set forth in the following description or illustrated in the drawings. The disclosed technology can be implemented in a variety of examples and can be practiced or carried out in various ways. In particular, the presently disclosed subject matter is described in the context of being a filter and filter assembly for filtering paint overspray particulates from air. The present disclosure, however, is not so limited, and can be applicable in other air filtering contexts. The present disclosure, for example and not limitation, can include filters and filter assemblies used in other types of air filtration systems. For example, and not limitation, the disclosed technology can be used in residential air filtration or other industrial and manufacturing air filtration systems (e.g., air filtration for metal working, wood working, composites finishing, foundries, etc.) or any other type of air filtration system used to filter particles from the air. Such implementations and applications are contemplated within the scope of the present disclosure. Accordingly, when the present disclosure is described in the context of being a filter and filter assembly for capturing paint overspray, it will be understood that other implementations can take the place of those referred to. Furthermore, when reference is made to air being the fluid that particulates are filtered from, it will be understood that other fluids can take the place of air depending on the particular application.

(15) It should also be noted that, as used in the specification and the appended claims, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. References to a composition containing “a” constituent is intended to include other constituents in addition to the one named.

(16) Also, in describing the examples, terminology will be resorted to for the sake of clarity. It is intended that each term contemplates its broadest meaning as understood by those skilled in the art and includes all technical equivalents which operate in a similar manner to accomplish a similar purpose.

(17) Herein, the use of terms such as “having,” “has,” “including,” or “includes” are open-ended and are intended to have the same meaning as terms such as “comprising” or “comprises” and not preclude the presence of other structure, material, or acts. Similarly, though the use of terms such as “can” or “may” are intended to be open-ended and to reflect that structure, material, or acts are not necessary, the failure to use such terms is not intended to reflect that structure, material, or acts are essential. To the extent that structure, material, or acts are presently considered to be essential, they are identified as such.

(18) The components described hereinafter as making up various elements of the disclosed technology are intended to be illustrative and not restrictive. Many suitable components that would perform the same or similar functions as the components described herein are intended to be embraced within the scope of the disclosed technology. Such other components not described herein can include, but are not limited to, for example, similar components that are developed after development of the presently disclosed subject matter.

(19) Referring now to the drawings, in which like numerals represent like elements, examples of the present disclosure are herein described. As will be described in greater detail, the present disclosure includes filters and filter assemblies for capturing particulates from air and, more particularly, to capturing paint overspray commonly resulting from use of aerosol paints and paint spray guns.

(20) As depicted in FIG. 1, the disclosed technology can include a filter media pack **100** having a filter layer **102**. The filter layer **102** can be a continuous piece of filter material and be configured to be folded along its length to a more compact configuration. For example, the filter media pack **100** can be folded to a compact configuration when shipped or stored and unfolded, or at least

partially unfolded, to be deployed as an air filter. The filter layer **102** can be made from any suitable type of filter media that is capable of filtering particulates from air or other fluids. For example, the filter layer **102** can be made from lofted or non-woven polyester, lofted or non-woven polypropylene, filament fiberglass, polyurethane foam, electrostatic polypropylene, shredded paper, cardboard, cotton, or any other suitable filter media for the particular application. No matter the material that the filter layer **102** is made from, the filter layer **102** can be configured to allow air (or other gases) to pass through the filter while particulates suspended in the air can be filtered from the air. Furthermore, the filter layer **102** can also be treated with a flame retarding agent, activated carbon, tackifier or other post-manufacturing treatment that can help to enhance the effectiveness and/or longevity of the filter layer **102**.

(21) The filter media pack **100** can include one or more seams **104** at various locations along the filter layer **102**. The seam **104**, for example, can be formed along an outer edge of the filter layer **102** and at locations where the filter layer **102** is likely to be bent either for storage or for use. The seam **104** can help to increase the durability of the filter layer **102**. For example, if the seam **104** is formed along an outer edge of the filter layer **102**, the seam **104** can help to keep the filter layer **102** together when, for example, the filter layer is made from lofted polyester. The seam **104** can be formed using any suitable means of forming the seam **104**. For example, the seam **104** can be formed by thermal bonding, stitching, glue, fasteners, or any other suitable manufacturing method of forming the seam **104**.

(22) FIG. 2A illustrates a top view of an example filter media pack **100**, in accordance with the disclosed technology. As depicted in FIG. 1, in some examples, the filter media pack **100** can include multiple filter layers to increase the effectiveness of the filter media pack **100**. For example, the filter media pack **100** can have a first filter layer **202** and a second filter layer **204** bonded together to form the filter media pack **100**. If the filter media pack **100** has more than a single filter layer, the filter media pack **100** can be made from multiple different types of filter media. For example, the first filter layer **202** can be made from a lofted polyester having a first density while the second filter layer **204** can be made from a lofted polyester having a second density. In other examples, the first filter layer **202** can be made from a lofted polyester while the second filter layer **204** can be made from fiberglass. As will be appreciated by one of skill in the art, the filter media pack **100** can be made from multiple layers of filter media being made from various materials and having various material properties to increase the effectiveness of the filter media pack **100** for the particular application.

(23) As depicted in FIG. 2A, the filter media pack **100** can include a filter media support **206** that can be bonded between the first filter layer **202** and the second filter layer **204**. The filter media support **206** can be configured to provide support to the first filter layer **202** and the second filter layer **204**. For example, the filter media support **206** can provide support and stability to the filter media pack **100** when the filter media pack **100** experiences a differential pressure caused by the air filtration system when the filter media pack **100** is used to filter particulates from air. The filter media support **206** can be made from supportive material such as cardboard, plastics, PVC, metals, composite materials, or any other suitable material for the application.

(24) In some examples, and as depicted in FIG. 2A, the filter media support **206** can be installed between the first filter layer **202** and the second filter layer **204** along portions of the filter media pack **100** intended to remain linear while the filter media support **206** can be absent from portions of the filter media pack **100** intended to be bent. By doing so, the filter media supports **206** can provide support to the filter media pack **100** along lengths of the filter media pack **100** that are likely to need the most support while also facilitating folding of the filter media pack **100** where the filter media supports **206** are absent. Additionally, because the filter media supports **206** can be made of a non-breathable material the filter media supports **206** can prevent a mixture of air and particulates from passing through the filter media pack **100** at locations where the filter media supports **206** are installed. Therefore, by not installing filter media supports **206** at various

locations of the filter media pack **100**, the filter media supports **206** can channel or direct the flow of a mixture of air and particulates through the filter media pack **100** at the locations where the filter media support is absent. For example, the filter media pack **100** depicted in FIG. 2A can permit the mixture of air and particulates to pass through the filter media pack **100** at the locations where the filter media support **206** is absent (e.g., at either end and each bend location of the filter media pack **100**). As will be appreciated by one of skill in the art, by causing the mixture of air and particulates to pass through the filter media pack **100** at specific locations, the filter media pack **100** can be configured to channel the flow of air and particulates such that the filter media pack **100** collects or captures particulates without prematurely clogging. For example, the filter media pack **100** can be configured with greater gaps between the filter media supports **206** at a back of the filter media pack **100** (nearer an outlet of the filter box described in greater detail herein) than gaps between the filter media supports **206** at the front of the filter media pack **100** (nearer an inlet of the filter box described in greater detail herein). In this way, the filter media pack **100** can be configured to channel or direct the flow of the air and particulates toward a back portion of the filter media pack **100** such that particulates are collected along the length of the filter media pack **100** rather than concentrated at a front portion of the filter media pack **100** which could lead to a premature clogging of the filter media pack **100**.

(25) FIGS. 2B and 2C illustrate side views of other example filter media supports **206**, in accordance with the disclosed technology. As depicted in FIG. 2B, the filter media support **206** can include apertures **208** that can help to channel a mixture of air and particulates through the filter media pack **100** at the locations of the apertures **208**. Similarly, as depicted in FIG. 2C, the filter media supports **206** can include slots **210** that can help to channel a mixture of air and particulates through the filter media pack at the locations of the slots **210**. The filter media supports **206** can be further configured such that a greater concentration of the apertures **208** or the slots **210** can be positioned nearer a back portion of the filter media pack **100** (or nearer an exit of the filter box described further herein) such that a greater portion of the mixture of air and particulates is directed along the length of the filter media pack **100** rather than concentrated at a front portion of the filter media pack **100** which could lead to a premature clogging of the filter media pack **100**.

Alternatively, if it is desirable to collect a greater amount of the particulates nearer a front portion of the filter media pack **100**, or a middle portion of the filter media pack **100**, the filter media supports **206** can have a greater concentration of the apertures **208** or the slots **210** near the front or middle portions of the filter media pack.

(26) The filter media pack **100** can be inserted into a filter housing **302** as depicted in FIG. 3 to form a filter box **300**. FIG. 3 illustrates a top view of an example filter box **300**, in accordance with the disclosed technology. The filter box **300** depicted in FIG. 3 is shown with a top of the filter box **300** opened such that the internal portions of the filter box **300** are visible. As depicted in FIG. 3, the filter housing **302** can receive and support the filter media pack **100** to form the filter box **300**. Furthermore, as depicted in FIG. 3, the air and particulate mix can be configured to enter a first side of the filter box **300** (e.g., at an inlet **301**) and exit the filter box **300** at an opposite side of the filter box **300** (e.g., at an outlet **303**). As the air and particulate mix passes through the filter media pack **100**, the particulates can be captured by, or adhere to, the filter media pack **100** such that filtered air can be released at the outlet **303**. The filtered air can be released to the atmosphere, recirculated back through the paint booth, or directed by the air filtration system through further filtering before being released to the atmosphere or recirculated back through the paint booth. Although depicted as having greater spaces between ends of the filter media pack **100** and a nearest bend of the filter media pack **100** than neighboring bends of the filter media pack, the filter housing **302** can be configured to have approximately equal spacing between the ends of the filter media pack **100** and each of the bends of the filter media pack **100**.

(27) The filter media pack **100** can be shaped in an alternating “V” wedge configuration (e.g., a zig-zag) inside of the filter housing **302**. However, the filter media pack **100** can be formed in many

different configurations to fit the particular application. As a non-limiting example, the filter media pack **100** can be folded over on itself during storage and shipment and then extended outward like an accordion when deployed in the filter housing **302** forming the alternating “V” wedge configuration. In some examples, the filter media pack **100** can be inserted into the filter housing **302** in the alternating “V” wedge configuration such that the widest part (i.e., the top portion of the “V”) faces the entering air side.

(28) To prevent air and particulates from escaping around the filter media pack **100** when the filter media pack **100** is assembled in the filter housing **302**, the filter media pack **100** can be sized slightly larger than the internal spacing of the filter housing **302** such that the filter media pack **100** is compressed or bent by the filter housing **302** to form a friction fit gasket **310**. In this way, the filter media pack **100** can form a tight seal along each of the edges of the filter media pack **100** that come into contact with the filter housing **302**. As will be appreciated by one of skill in the art, this can include the top and bottom portions of the filter media pack **100** contacting a top and bottom portion of the filter housing **302** (i.e., the portions of the filter media pack **100** and filter housing **302** that are positionally located into and out of the plane (or page) of FIG. 3) as well as the portions of the filter media pack **100** that contact a front and back portion of the filter housing **302** (i.e., the upper and lower portions of the filter media pack **100** and the filter housing **302** as depicted in FIG. 3).

(29) The filter housing **302** can include one or more portions that are configured to support and align the filter media pack **100** inside of the filter housing **302**. For example, the filter housing **302** can include inlet guide fingers **304**, inlet support tabs **306**, outlet guide fingers **307**, and outlet support tabs **308** (collectively, “the Support Tabs”). As will be explained in greater detail herein, the Support Tabs can each be part of the material that makes up the filter housing **302** and can be configured to fold inwardly from an outer surface of the filter housing **302**. The Support Tabs can each be sized and positioned such that each of the Support Tabs contact an edge of the filter media pack **100** and help to support the filter media pack **100** when a force is applied to the filter media pack **100** (e.g., when a pressure differential is applied to the filter media pack **100** from an air filtration system to cause the mixture of air and particulates to pass through the filter media pack **100**). In this way, the Support Tabs can be configured to prevent movement of the filter media pack **100** when a force is applied to the filter media pack **100** and ensure the mixture of air and particulates passes through the filter media pack **100** rather than around it.

(30) The inlet guide fingers **304** can be sized and positioned such that the inlet guide fingers **304** contact an edge of a front surface (i.e., the surface facing the inlet **301**) of the filter media pack **100** near bends in the middle portion of the filter media pack and help to support the filter media pack **100** when a force is applied to the filter media pack **100**. The inlet guide fingers **304** can be sized such that the filter media pack **100** can contact the inlet guide fingers **304** along an edge of the inlet guide fingers **304**. Furthermore, the inlet guide fingers **304** can be sized and positioned to ensure the filter media pack **100** is aligned or positioned properly within the filter housing **302**.

(31) The inlet support tabs **306** can be sized and positioned such that the inlet support tabs **306** can fold inwardly and contact the filter media pack **100** near an end of the filter media pack **100**. The filter media pack **100** and the filter housing **302** can be configured such that the ends of the filter media pack **100** can be tucked behind an edge of the filter housing **302** proximate the inlet **301** and the inlet support tabs **306** can help to keep the ends of the filter media pack **100** tucked behind the edge of the filter housing **302**. In this way, the inlet support tabs **306** can help to ensure the filter media pack **100** is pressed against an edge of the filter housing **302** such that the filter media pack **100** remains in place and the mixture of air and particulates is prevented from escaping around the filter media pack **100**.

(32) The outlet guide fingers **307** can be sized and positioned such that the outlet guide fingers **307** contact an edge of a rear surface (i.e., the surface facing the outlet **303**) of the filter media pack **100** near bends in the middle portion of the filter media pack and help to support the filter media pack

100 when a force is applied to the filter media pack **100**. The outlet guide fingers **307** can be sized such that the filter media pack **100** can contact the outlet guide fingers **307** along a length of the outlet guide fingers **307**. Furthermore, similar to the inlet guide fingers **304**, the outlet guide fingers **307** can be sized and positioned to ensure the filter media pack **100** is aligned or positioned properly within the filter housing **302**.

(33) The outlet support tabs **308** can be sized and positioned such that the outlet support tabs **308** can fold inwardly and contact the rear surface (i.e., the surface facing the outlet **303**) of the filter media pack **100** near outer ends of bends in the middle portion of the filter media pack **100**. The outlet support tabs **308** can help to support the filter media pack **100** when a force is applied to the filter media pack **100**.

(34) FIG. 4A illustrates a front perspective view of the filter box **300**, in accordance with the disclosed technology. As depicted in FIG. 4A, the filter housing **302** can enclose the filter media pack **100** to form the filter box **300**. The inlet **301** can be formed from material forming a wall of the filter box **300** and can be sized and positioned to permit a mixture of air and particulates to pass through the filter box **300** to create filtered air by passing through the filter media pack **100**.

(35) As depicted in 4A, and as will be described in greater detail herein, the inlet guide fingers **304** can include inlet guide finger perforations **404** that can facilitate the inlet guide fingers **304** being folded inwardly into the filter housing **302** to form the inlet **301** and position the inlet guide fingers **304** to engage the filter media pack **100**. Similarly, the inlet support tabs **306** can include inlet support tab perforations **406** that can facilitate the inlet support tabs **306** being folded inwardly into the filter housing **302** to form the inlet **301** and position the inlet support tabs **306** to engage the filter media pack **100**. The inlet guide finger perforations **404** and the inlet support tab perforations **406** can be sized such that a user can push the inlet guide fingers **304** and the inlet support tabs **306** inwardly without tearing the filter housing **302**.

(36) FIG. 4B illustrates a rear perspective view of the filter box **300**, in accordance with the disclosed technology. As depicted in FIG. 4B, the filter housing **302** can enclose the filter media pack **100** to form the filter box **300**. The outlet **303** can be formed from material forming a wall of the filter box **300** and can be sized and positioned to permit a mixture of air and particulates to pass through the filter box **300** to create filtered air by passing through the filter media pack **100**.

(37) As depicted in 4B, and as will be described in greater detail herein, the outlet guide fingers **307** can include outlet guide finger perforations **407** that can facilitate the outlet guide fingers **307** being folded inwardly into the filter housing **302** to form the outlet **303** and position the outlet guide fingers **307** to engage the filter media pack **100**. Similarly, the outlet support tabs **308** can include outlet support tab perforations **408** that can facilitate the outlet support tabs **308** being folded inwardly into the filter housing **302** to form the outlet **303** and position the outlet support tabs **308** to engage the filter media pack **100**. The outlet guide finger perforations **407** and the outlet support tab perforations **408** can be sized such that a user can push the outlet guide fingers **307** and the outlet support tabs **308** inwardly without tearing the filter housing **302**.

(38) FIG. 5A illustrates a front view of a filter box **300**, in accordance with the disclosed technology. The filter box **300** illustrated in FIG. 5A depicts the inlet guide fingers **304** and the inlet support tabs **306** before the inlet guide fingers **304** and the inlet support tabs **306** are pushed inwardly to form the inlet **301**. As depicted, the inlet guide fingers **304** can each include at least two sets of inlet guide finger perforations **404** to facilitate the inlet guide fingers **304** being pushed inwardly to form the inlet **301**. By including at least two sets of inlet guide finger perforations **404**, the inlet guide fingers **304** can be better able to engage the filter media pack **100** while allowing greater spacing for the inlet **301**. The inlet support tabs **306** can include at least one set of inlet support tab perforations **406** such that the inlet support tabs **306** can be pushed inwardly and engage the filter media pack **100** near the ends of the filter media pack **100** to help keep the filter media pack **100** in place.

(39) FIG. 5B illustrates a front view of a filter box **300**, in accordance with the disclosed

technology. Similar to FIG. 5A, the filter box **300** illustrated in FIG. 5B depicts the outlet guide fingers **307** and the outlet support tabs **308** before the outlet guide fingers **307** and the outlet support tabs **308** are pushed inwardly to form the inlet **301**. As depicted, the outlet guide fingers **307** can each include at least two sets of outlet guide finger perforations **407** to facilitate the outlet guide fingers **307** being pushed inwardly to form the outlet **303**. By including at least two sets of outlet guide finger perforations **407**, the outlet guide fingers **307** can be better able to engage the filter media pack **100** while allowing greater spacing for the outlet **303**.

(40) The outlet support tabs **308** can include outlet support tab extensions **508** that can be folded inwardly and horizontally to engage the filter media pack **100**. The outlet support tab extensions **508** can be formed by the outlet support tabs **308** including at least three sets of outlet support tab perforations **408**—one of which can facilitate the outlet support tabs **308** being pushed inwardly into the filter box **300** while the other two outlet support tab perforations **408** can facilitate the outlet support tab extensions **508** being oriented horizontally to engage and help to support the filter media pack **100**.

(41) FIG. 6 illustrates a top view of an example filter housing **302** in an unassembled state, in accordance with the disclosed technology. As depicted in FIG. 6, in some examples the filter housing **302** can be made from a single piece of material. For example, the filter housing **302** can be made from a single piece of cardboard and be configured such that the filter housing **302** can be folded from an unassembled state (as depicted in FIG. 6) to the assembled state (as depicted in FIGS. 4A-5B). In this way, the filter housing **302** can be configured to be generally flat when in an unassembled state which can require less space when the filter housing **302** is stored or shipped. Furthermore, as will be appreciated by one of skill in the art, if the filter housing **302** is made from a single piece of cardboard, the filter housing **302** can be made using a cutting die, a cutting machine, or other similar manufacturing processes for manufacturing cardboard boxes.

Alternatively, the filter housing **302** can be made from multiple pieces of material affixed to each other to form the filter housing **302**.

(42) The filter box **300**, as described previously, can be installed in a location where a mixture of air and particulates is directed from a paint booth to ensure particulates in the air are filtered from the air prior to the air being released to the environment or recycled back through the paint booth. A single filter box **300** can be used for smaller filtration applications and multiple filter boxes **300** can be used for larger filtration applications. Furthermore, the size and shape of the filter box **300** can be varied as appropriate for the particular application. For example, the filter box **300** can be smaller for air filtration systems having a relatively low filtration capacity and larger for air filtration systems having a relatively high filtration capacity.

(43) In some examples, more than one filter box **300** can be assembled together to form a filter assembly **700** as depicted in FIG. 7. The filter assembly **700** can be installed in an air filtration system for filtering particulates from a mixture of particulates and air. For example, the filter assembly **700** can be formed by multiple filter boxes **300** being placed within a filter assembly housing **702** to create a filter assembly **700** to be installed in an air filtration system. As a specific example, and as depicted in FIG. 7, multiple filter boxes **300** can be stacked vertically within a filter assembly housing **702** configured to direct air from the filter assembly inlet **706** located at a top portion of the filter assembly housing **702**, through the filter boxes **300**, and out a filter assembly outlet **707** located at a side portion of the filter assembly housing **702**. This configuration can be, for example, installed in air filtration systems that utilize a downdraft in a paint booth to capture particulates. The filter assembly **700** can be sized to fit existing air filtration systems such that a user can simply assemble the filter assembly **700**, remove a used filter assembly **700**, and replace the used filter assembly **700** with the newly assembled filter assembly **700**. Depending on the application, replacing used filter assemblies **700** with new filter assemblies **700** can be accomplished when the air filtration system is shut down or while the air filtration system is still in operation.

(44) In some examples, the filter assembly housing **702** for holding the multiple filter boxes **300** can be made from the same material used to make the supportive exterior of the filter box (e.g., cardboard, plastic, PVC, metals, composite materials, or any other suitable material for the application). The filter assembly housing **702** can be configured to form a nearly air-tight barrier around the filter boxes **300** such that the mixture of air and particulates is directed through the filter boxes **300** and not around them. Furthermore, the filter assembly housing **702** can be configured to have spacers and supports in predetermined locations such that the filter assembly housing **702** is able to help ensure the filter boxes **300** are properly spaced and supported when installed in the filter assembly housing **702**. The filter assembly housing **702** can also include reinforcing material located where the filter assembly housing **702** is likely to be lifted and secured in the air filtration system. For example, the filter assembly housing **702** can have reinforcing tabs **714** located where a restraint device (e.g., pneumatic clamp, hydraulic clamp, manual clamp, or other device configured to keep the filter assembly **700** in place) is likely to grip the filter assembly housing **702** to hold the filter assembly **700** in place. Furthermore, the filter assembly housing **702** can be mounted on a base structure **704** that can be used to lift and move the filter assembly **700** with, for example, a pallet jack or forklift. For example, the base structure **704** can be a pallet, a wooden base, a reinforced cardboard base, a plastic base, or any other suitable type of base structure for the application.

(45) If the filter boxes **300** are installed vertically together into a filter assembly housing **702** to form a filter assembly **700**, the filter boxes can be offset horizontally, as depicted in FIG. 7, to facilitate improved air flow during use. As one of skill in the art will appreciate, under certain operating conditions, the filter box **300** nearest the filter assembly inlet **706** is likely to be the first to clog during use because a greater number of particulates will initially be directed through the filter box **300** nearest the filter assembly inlet **706**. Furthermore, as particulates are collected on the filter box **300** nearest the filter assembly inlet **706**, the particulates will build up and begin to restrict air flow through the filter box **300**. If the particulates continue to build up, the passage between the filter box **300** and the filter assembly housing **702** can eventually be restricted and air flow reduced. Thus, by offsetting the filter boxes **300** horizontally such that the largest clearance between the filter box **300** and the filter assembly housing **702** is nearest the filter assembly inlet **706** and the smallest clearance between the filter box **300** and the filter assembly housing **702** is farthest from the filter assembly inlet **706**, the mixture of air and particulates is able to reach the farthest filter box **300** from the filter assembly inlet **706** as the other filter boxes **300** become clogged during use. In other examples, due to comparatively higher operating air velocities, a filter box **300** further from the inlet **706** can clog during use before the other filter boxes **300**. In this situation, particulates begin to collect on the remaining filter boxes **300** as the air and particulates are directed around the clogged filter box **300** and toward remaining unclogged filter boxes **300**.

(46) The filter assembly **700** can include a sloped edge **708** at the filter assembly inlet **706** air deflectors **710** between each filter box **300** to help improve air flow through the filter assembly **700**. For example, as depicted in FIG. 7, the inlet can include a sloped edge **708** to help direct the air and particulates through the filter assembly inlet **706** and toward the filter boxes **300** as the mixture of air and particulates enter the filter assembly **700**. The sloped edge **708** near the filter assembly inlet **706** can be sized and shaped as suitable for the particular application. Furthermore, as depicted in FIG. 7, the filter assembly **700** can include air deflectors **710** between each filter box **300** to eliminate a flat edge of a top portion of a filter box **300** resulting from the offset between the filter boxes **300** that could otherwise cause a greater pressure differential as air is directed through the filter assembly **700**. The sloped edge **708** and the air deflectors **710** can be made from the same material as the filter assembly housing **702** (e.g., cardboard, plastic, PVC, metals, composite materials, or any other suitable material for the application) or the sloped edge **708** and the air deflectors **710** can be made from material that is different than the filter assembly housing **702** depending on the application.

(47) The filter assembly **700** can further include a supplemental filter **712** that can be installed proximate the filter assembly outlet **707** to filter particulates that may escape past the filter boxes **300**. For example, the supplemental filter **712** can be a sheet of filter media stretched across the filter assembly outlet **707**. The supplemental filter **712** can be made from the same material as the filter layer **102** of the supplemental filter **712** can be made from material that is different than the filter layer **102**.

(48) The filter assembly **700** just described and depicted in FIG. 7 is offered only for illustrative purposes and should not be construed as limiting. As further non-limiting illustrative examples, the filter boxes **300** can be stacked horizontally (side-by-side) and placed in a filter assembly housing **702** for applications where it would be advantageous to filter the mixture of air and particulates that enters and exits the filter assembly **700** horizontally. In other examples, the filter boxes **300** can be positioned vertically (installed with the inlet **301** facing upwards and the outlet **303** downwards) for applications where it would be advantageous to filter a mixture of air and particulates that enters and exits the filter assembly **700** vertically. In yet other examples, the filter boxes **300** can be installed in series (one filter box **300** placed directly after another filter box **300**) for applications where it would be advantageous to provide additional filtering. As one of skill in the art will appreciate, the disclosed technology can comprise many different configurations to fit a particular application. In each configuration, many of the features previously described can be included such as each filter box **300** being offset from an adjacent filter box **300**, a sloped edge **708**, air deflectors **710**, etc. Furthermore, the size and shape of the filter assembly **700** can be varied as appropriate for the particular application. For example, the filter assembly **700** can be smaller for air filtration systems having a relatively low filtration capacity and larger for air filtration systems having a relatively high filtration capacity. As a non-limiting example, the filter assembly **700** can have dimensions of 85 inches tall, 32.5 inches wide, and 48.375 inches depth.

(49) While the present disclosure has been described in connection with a plurality of exemplary aspects, as illustrated in the various figures and discussed above, it is understood that other similar aspects can be used, or modifications and additions can be made to the described aspects for performing the same function of the present disclosure without deviating therefrom. For example, in various aspects of the disclosure, methods and compositions were described according to aspects of the presently disclosed subject matter. But other equivalent methods or composition to these described aspects are also contemplated by the teachings herein. Therefore, the present disclosure should not be limited to any single aspect, but rather construed in breadth and scope in accordance with the appended claims.

Claims

1. An air filter for filtering particulates from a mixture of air and particulates, the air filter comprising: a filter housing comprising a single planar sheet of cardboard configured to be folded from a relatively flat state to a deployed state, the filter housing comprising an inlet and an outlet; and a filter media pack disposed at least partially within the filter housing and configured to receive a mixture of air and particulates through the inlet and filter at least a portion of the particulates from the mixture of air and particulates to create filtered air directed through the outlet, at least a portion of the filter media pack extending at an oblique angle with respect to a direction extending from the inlet to the outlet, the filter media pack comprising: a filter layer comprising one or more filter sections; and a plurality of filter media supports disposed along the filter layer from approximately the inlet to approximately the outlet, the plurality of filter media supports being configured to support the filter layer, wherein a first filter media support of the plurality of filter media supports is spaced a distance apart from a second filter media support of the plurality of filter media supports nearest the inlet.

2. The air filter of claim 1, wherein the filter media pack is configured to be folded and placed

inside of the filter housing for cooperative engagement with the filter housing.

3. The air filter of claim 2, wherein the filter media pack is configured to form a friction fit seal at an interface between the filter media pack and the filter housing, wherein the friction fit seal is configured to prevent the mixture of air and particulates from being directed around an outer edge of the filter media pack.

4. The air filter of claim 1, wherein at least one filter media support of the plurality of filter media supports comprises one or more openings configured to channel the mixture of air and particulates through the filter layer.

5. The air filter of claim 4, wherein the filter layer comprises a first filter layer, the filter media pack further comprising a second filter layer.

6. The air filter of claim 5, wherein at least one filter media support of the plurality of filter media supports is disposed at least partially between the first filter layer and the second filter layer.

7. The air filter of claim 1, wherein filter housing further comprises perforations at least partially defining the inlet and the outlet, the perforations configured to facilitate removal of the cardboard from the single planar sheet of cardboard to form the inlet and the outlet.

8. The air filter of claim 7, wherein the perforations further define one or more support tabs configured to be folded inwardly from at least the inlet to support the filter media pack inside of the filter housing.

9. The air filter of claim 1, wherein a third filter media support of the plurality of filter media supports is spaced a distance apart from a fourth filter media support of the plurality of filter media supports nearest the outlet.

10. The air filter of claim 9, wherein the distance between the first filter media support and the second filter media support is greater than the distance between the third filter media support and the fourth filter media support.

11. An air filter for filtering particulates from a mixture of air and particulates, the air filter comprising: a filter housing comprising a single planar sheet of cardboard configured to be folded from a relatively flat state to a deployed state, the filter housing comprising a plurality of perforations at least partially defining an inlet and an outlet, the perforations configured to facilitate removal of at least some of the cardboard from the single planar sheet of cardboard to form the inlet and the outlet; and a filter media pack disposed at least partially within the filter housing and configured to receive a mixture of air and particulates through the inlet and filter at least a portion of the particulates from the mixture of air and particulates to create filtered air directed through the outlet, the filter media pack comprising: a filter layer comprising one or more filter sections; and a plurality of filter media supports disposed along the filter layer from approximately the inlet to approximately the outlet, the plurality of filter media supports being configured to support the filter layer, wherein a first filter media support of the plurality of filter media supports is spaced a distance apart from a second filter media support of the plurality of filter media supports nearest the inlet.

12. The air filter of claim 11, wherein the perforations further define one or more support tabs configured to be folded inwardly from at least the inlet to support the filter media pack inside of the filter housing.

13. The air filter of claim 12, wherein the one or more support tabs comprise an inlet guide finger configured to fold inwardly from a top or a bottom of a front outer wall of the filter housing to, at least partially, form the inlet.

14. The air filter of claim 12, wherein the one or more support tabs comprise an inlet support tab configured to fold inwardly from at least one side of a front outer wall of the filter housing to, at least partially, form the inlet.

15. The air filter of claim 12, wherein the one or more support tabs comprise an outlet guide finger configured to fold inwardly from a top or a bottom of a rear outer wall of the filter housing to, at least partially, form the outlet.

16. The air filter of claim 12, wherein the one or more support tabs comprise an outlet support tab configured to fold inwardly from at least one side of a rear outer wall of the filter housing to, at least partially, form the outlet.

17. An air filter for filtering particulates from a mixture of air and particulates, the air filter comprising: a filter media pack configured to receive a mixture of air and particulates and filter at least a portion of the particulates from the mixture of air and particulates to create filtered air, the filter media pack comprising: a filter layer comprising one or more filter sections; and a plurality of filter media supports configured to support the filter layer; and a filter housing configured to receive and support the filter media pack, the filter housing comprising a material configured to be folded from a relatively flat state to a deployed state and comprising a plurality of perforations at least partially defining an inlet and an outlet, the perforations configured to facilitate removal of at least a portion of the material to form the inlet and the outlet, wherein the air filter is configured to channel the mixture of air and particulates from the inlet, through the filter media pack, and out the outlet to create the filtered air, and wherein the plurality of filter media supports is disposed along the filter layer from approximately the inlet to approximately the outlet and a first filter media support of the plurality of filter media supports is spaced a distance apart from a second filter media support of the plurality of filter media supports nearest the outlet.

18. The air filter of claim 17, wherein at least a portion of the filter media pack extends at an angle with respect to a direction extending from the inlet to the outlet.

19. The air filter of claim 17, wherein at least one filter media support of the filter plurality of media supports comprises one or more openings configured to channel the mixture of air and particulates through the filter layer.

20. The air filter of claim 19, wherein the filter layer comprises a first filter layer, the filter media pack further comprising a second filter layer, the filter media support being disposed at least partially between the first filter layer and the second filter layer.
